Journal of Psychological Inquiry
2025, Vol. 29, No. 2, pp. 62-71
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DO THE SEMANTIC ASSOCIATIONS OF VISUAL ITEMS INFLUENCE VISUAL SEARCH PERFORMANCE?

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Abstract – In traditional visual search experiments, participants search through displays that contain a single target item and varying numbers of nontarget distractors. Targets and distractors had traditionally been defined by distinct visual features such as their color and shape, but eventually researchers began to wonder whether search items' semantic associations, such as an A that is a letter and an 8 that is a number, could affect search performance. Although the initial attempt to do so demonstrated that category membership (i.e., numbers versus letters) influenced visual search, repeated failures to replicate the initial effect discouraged further attempts for the next two decades. One major problem with such studies was that any effects of semantic associations could also be explained as resulting from shape differences: an A is in a different category than an 8, but they also have different shapes. A breakthrough study managed to find that visual search was more efficient when the distractors' shapes had the same semantic identity (i.e., B and b) than when they had different identities (i.e., B and p), while controlling for the search items' shapes. The author of this breakthrough argued that his results undermined the classical perception-cognition divide, whereby higher level cognitive processes such as knowledge and expectation have traditionally been presumed not to affect basic perceptual processing. Nonetheless, in this review we argue that this study as well as later studies that built on its success can be interpreted to be consistent with the classical perception-cognition divide. Specifically, we argue that search items' semantic associations affect the role of cognitive processing in visual search rather than the basic perceptual processing.

Keywords: visual search, semantic associations

Much of life is spent searching for things: looking for your car in a mall parking lot, searching for a friend's familiar face in a crowded room, or trying to find your favorite cereal in the packed grocery store aisle. In each of these scenarios there are several items to search through and just one item that is the target, whether that be a car, a friend, or a box of Cheerios. What guides visual searches for these targets? How do people process what they are seeing and discover where the target is? A reasonable first guess might claim that search begins by picking one visual location at random, then checking to see if the target is at the chosen location, and if not, randomly selecting another location. But the visual search literature shows that searches rely on more efficient strategies than this kind of random selection. When looking for a specific target, searchers begin with the target's defined visual features, and limit the range of search to just the items that have that feature (Bacon & Egeth, 1997). Distractor items that do not have the target feature are eliminated from consideration. By using target features to limit the range of search items, these features are said to be guiding features (Wolfe, 1998).

This reflects the long-held view that visual search performance is generally presumed to be influenced by the visual features of the visual display items, and that any meanings associated with those items have no bearing on search performance. Indeed, in the seminal paper that laid the groundwork for the bulk of the visual search literature (Treisman & Gelade, 1980), the target letter T is distinct from the distractor letter X due to their shapes, rather than the fact that they have different roles in written language. The notion that perceptual and cognitive processing operate independently was later formalized (Pylyshyn, 1999; Firestone & Scholl, 2015) as the perception-cognition divide. The perception-cognition divide asserts that while perception can

undoubtedly influence aspects of cognition such as thoughts and beliefs, the information flows in just one direction from perception to cognition but not the other direction, because thoughts and beliefs cannot alter the way the world looks, sounds, and feels. Cognition can play a role in the way we interpret perceptions, but can't alter basic perceptual appearances. This idea of the perception-cognition divide was later applied in the context of visual search, by Wolfe and Horowitz (2004), who reviewed the visual search literature to find what attributes influence visual search performance. The authors found that visual features such as color and shape definitely guide search. Previous attempts to find effects of semantic associations, however, such as whether an item was a letter versus a number, had been unsuccessful, leading Wolfe and Horowitz to doubt that these kinds of attributes guide search. The distinction between visual attributes that definitely guide search and semantic attributes that are doubted to guide search supports the idea of the perception-cognition divide; color and shape work at the perceptual level and semantic associations work at the cognitive level.

The arguments presented by Wolfe and Horowitz (2004) suggested that any attempts to find an effect of semantic associations on visual search would be misguided. Nevertheless, Lupyan (2008) took their conclusions as a challenge, and found a new way to reveal the role of semantic associations in visual search, thereby opening the door for others to find new ways in which semantic associations might influence visual search. In this review, we begin with early attempts that failed to find any effect of semantic associations on visual search, then continue with Lupyan's groundbreaking work that opened the door, then highlight the work of other researchers who built on the foundation he laid. For example, in searches that use numerical targets, the distance on the number line between the target's and distractors' positions affects visual search efficiency, as does the congruity between the target's numerical and physical size (i.e., font size). The Stroop effect, in which incongruities of word meaning and color slow responses, had a similar effect on visual search performance. These effects demonstrate that semantic associations *do* influence visual search. Nevertheless, we will argue that all of these results can be interpreted as consistent with the perception-cognition divide, because semantic associations influence mechanisms such as working memory that inhabit the cognitive side of the divide.

Visual Features Versus Semantic Associations

In their review of the kinds of features that can guide search, Wolfe and Horowitz (2004) argued that color and shape are undoubtedly visual features that can be used to guide search, a conclusion backed by decades of research. During visual search a guiding feature is used in the searching process to distinguish a target item from nontarget distractor items. For example, if you were partial to green Skittles candy, you would like to pick some out from a bowl filled with M&Ms and Skittles all mixed together. The green skittle is your target item, so you could start your search by removing all the candies that are not green from consideration. Human visual systems naturally have this ability to differentiate colors in search. Once you reduce the range of items to just those with the target color, locating the items with the target color that are labeled with an "S" shape would lead you to the target. Studies involving undoubted attributes such as color and shape have shown response time (RT) patterns indicating that participants can limit their search to just the items that display a target color or shape while ignoring the items that have a different color and shape (Egeth et al., 1984).

In contrast to visual features like color and shape, Wolfe and Horowitz (2004) argued that the semantic associations of a search item are unlikely to be useful in guiding search. Semantic associations refer to when a word or symbol has a meaning assigned to it, such as the numeral 4 that has a distinct shape, but also has an associated meaning. Students learn in grade school that each numeral has its own numerical size and corresponding location on a number line. Suppose participants were asked to search for a 7 or an 8 among several 3s, and if performance is different for 7 compared to 8, it would be tempting to conclude that the difference is attributable to numerical size, such that a 7 is closer to the 3s on the number line than 8. But Wolfe and Horowitz argued that there are two problems with this conclusion. First, while the human visual system can readily distinguish items on the basis of their color and shape, it is unimaginable that innate visual processing mechanisms would organize visual items by numerical size. Second, the 7 has a shape that is much more distinct from the shape of the 3s relative to 8, so any effect of numerical size could be driven by the visual feature of shape rather than semantic associations. Wolfe and Horowitz based their argument on experiments that tried to find effects of semantic associations on visual search, which began with Jonides and Gleitman (1972).

In an early attempt to investigate whether semantic associations can affect visual search performance, Jonides and Gleitman (1972) asked participants to search for a circle target; in one condition, participants were instructed to search for the "zero", and in another condition to search for the "oh". A zero is a number and an oh is a letter, so they belong to different

semantic categories even though they look the same. The circular target was presented among various numbers of nontarget distractors that were either in the same semantic category or a different category. For example, an oh letter target among the numbers 5, 6, 8 and 9 would be different target and distractor categories, whereas a zero number target would be the same category as these four number distractors. When the targets and distractors were from the same semantic category, RTs increased with the number of search items, so in other words responses were slower when searching among six items than when searching among two items, but when targets and distractors were from different categories, RTs were flat across increasing numbers of search items. This distinction between steeply increasing RTs as a function of the number of search items versus flat RT is a classic way to indicate the efficiency of search (Wolfe, 1998), where flat RT functions indicate relatively efficient search compared with tasks that elicit steep RT functions. These results, in which the RT functions are steeper when targets and distractors are the same category is analogous to the role of color in search: when target and distractor colors are similar, RT functions are steeper than when target and distractor colors are distinct (D'Zmura, 1991). The results from Jonides and Gleitman provided early support for the argument that semantic associations can affect visual search performance.

Duncan (1983) tried and failed to replicate the Jonides and Gleitman (1972) oh-zero effect in his own experiment. In Duncan's attempt to replicate Jonides and Gleitman's effect, he designed an experiment using the same set of stimuli and instructions. Distractor items were either letters or numbers and participants were instructed that the target was either an oh or zero. Duncan's results led him to conclude that there is no effect from different target and distractor categories. Even though the number zero and letter oh are from different categories, Duncan argued that the visual feature of shape is more important in visual search than the semantic associations of category membership.

Krueger (1984) built on Duncan's (1983) failure to replicate Jonides and Gleitman (1972) by expanding the target and distractor sets, and controlling the shapes of the items in the numerical and alphabetical categories. Participants were instructed to search for a target that was a 5, 6, S, or G. The target shapes were designed so the 5 was similar to the S, and the 6 was similar to the G. The numerical distractors were 1, 2, 3, 4, 7, 9 and the letter distractors were L, Z, B, K, J or P. As with the targets, the distractors were matched for appearance so the 1 and L were similar, as was the 2 and Z, 3 and B, 4 and K, 7 and J, and 9 and P. Krueger found that when letters and digits had physically similar appearances, the target being in a different category from distractors (e.g., a 5 target among Z, B, K, J, and P distractors) did not lead to an advantage in RTs. These results supported Duncan's argument that distinguishable shapes rather than category membership drives visual search performance. Taken together, these studies (Duncan, 1983; Jonides & Gleitman, 1972; Krueger, 1984) support the idea of the perception-cognition divide. Jonides and Gleitman (1972) provided early evidence challenging the perception-cognition divide, but then both Duncan (1983) and Krueger (1984) failed to replicate these findings, suggesting that cognitive processing in the form of semantic associations has no role in visual search performance.

Breakthrough: Letters That Are the Same or Different

Based on the results of Duncan (1983) and Krueger (1984), Wolfe and Horowitz (2004) argued that the ability for semantic associations of alphanumeric characters to guide search is doubtful. It seemed as though the question was settled: the semantic associations of alphanumeric characters do not guide search. This conclusion is consistent with the classical perception-cognition divide. As described by Pylyshyn (1999), higher level cognitive processes such as knowledge and expectation do not influence lower level perceptual processing, or in other words, perception is cognitively impenetrable. But where Pylyshyn provided evidence about numerous different kinds of visual processing, Wolfe and Horowitz's claims are specific to visual search performance.

Lupyan (2008) was not satisfied with this consensus and worked to reintroduce the idea of semantic associations having a direct effect on visual processing. The results from Duncan (1983) and Krueger (1984) showed that category membership was unlikely to affect search performance, so Lupyan used search items whose semantic similarity was more specific than category membership. Furthermore, Lupyan knew that he needed to carefully control the shapes of the search items. To do so, Lupyan had participants search for a character whose shape looks to modern eyes like a blend of a "b" and "p", but in the Old English alphabet, it was called the thorn (þ). Most modern English readers do not recognize the thorn, but the thorn's visual similarities to both letters b and p could be expected to activate participants' visual memories for both letters. Participants were instructed to localize the þ target from among (upper case) B and (lower case) p nontarget distractors in one condition, and among (upper case) B and (lower case) b distractors in the other condition. In the B and p condition, the distractors were not just two

different shapes but were also semantically distinct because the B and p are two different letters, but in the B and b condition, the distractors were the same letter. Visual search was less efficient when the distractors were two different letters (i.e., B and p) than when distractors were the same letter (i.e., B and b). By using specific shapes, rather than general categories, and controlling the shapes of the target and distractors, Lupyan managed to find an effect of semantic associations on visual search where Duncan (1983) and Krueger (1984) had failed. Further, he argued that the conceptual relationship between distractors (i.e., different letter identities for B and p versus the same identity for B and b) penetrated the perceptual processing needed to carry out the search task. That is, Lupyan argued that his results undermine Pylyshyn's argument that perception is cognitively impenetrable.

However, an alternative explanation of Lupyan's (2008) results that is consistent with Pylyshyn's (1999) claim that perception is cognitively impenetrable (later amplified by Firestone & Scholl, 2015), is that semantic associations affect the role of working memory in visual search (Kane et al., 2006) rather than visual processing per se. According to the working memory account, when search items such as those in Lupyan's experiment occupy the same conceptual identity in working memory (i.e., B and b), the mental representation is simpler than when they occupy different conceptual identities (i.e., B and p). As a result, the semantic similarity between items affects search efficiency because remembering what distractors need to be ignored requires less mental effort when the distractors occupy the same identity (B and b) than when they occupy different identities (B and p). Thus, Lupyan's results are consistent with both the cognitive penetrability and working memory accounts. While the interpretation of Lupyan's results is debatable, his primary contribution to the visual search literature was to reopen the door that had been closed by Duncan (1983) and Krueger (1984), which encouraged other experimenters to look for effects of semantic associations on visual search. Later studies built on Lupyan's breakthrough by looking at how numerical size affects visual search performance.

The Effect of Number Line Position

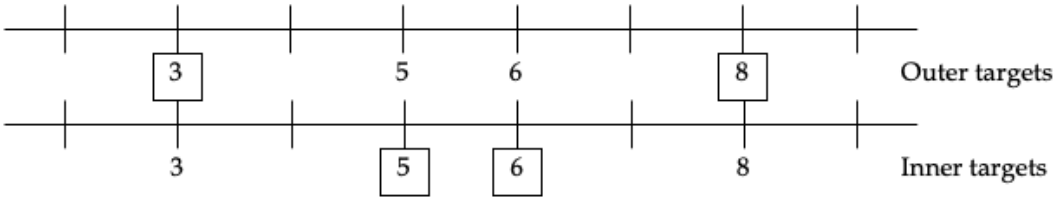
Schwarz and Eiselt (2012) were the first to look for effects of numerical distance on visual search efficiency. They varied the numerical distance between targets and distractors across visual displays. In an attempt to control the influence of shape on participants' search in the experiments, Schwarz and Eiselt used the digit 5 as a target in all their displays. They found that search was more efficient when the mean numerical distance between the target digit (5) and distractor digits was large (e.g., a 5 target among distractor digits 1, 2, 8, and 9) than when target-distractor numerical distance was small (e.g., a 5 target among distractor digits 3, 4, 6, and 7). This influence of number line position on visual search is equivalent to the influence of visual features such as color and shape, in which search efficiency increases with the target-distractor visual distinctiveness (Duncan & Humphreys, 1989). One limitation of the Schwarz and Eiselt study is that the target was always the same in all displays. Although the decision to use a single target item allowed Schwarz and Eiselt to control the target shape, their results may be specific to the chosen target digit of 5 and may not generalize to other target digits. Furthermore, the distractors that were numerically close to the target on the number line had different shapes than the numerically distant distractors, so the role of shape remains a plausible explanation for their results. Later studies looking at visual search for numerals developed further techniques to control for shape while broadening the target set to include more than one digit.

Godwin et al. (2014) expanded on the work of Schwarz and Eiselt (2012) by using multidimensional scaling to create a two-dimensional map of visual similarity between all the digits from 0 to 9. In the initial phase of their study, participants ranked the similarity in shape between every pair of digits in a Verdana font, and the researchers generated a two-dimensional map in which visually similar items (e.g., 2 and 5) were closer to each other on the map than visually distinct items (e.g., 4 and 8), which were farther away from each other. Godwin et al. thus had two maps, the one they built depicting visual similarity in two-dimensions, and the one-dimensional mental number line that most people develop in the course of their basic mathematical education (Pinhas et al., 2013). Thus equipped with a visual similarity and numerical similarity mental maps, Godwin et al. were able to tease apart the effects of visual and numerical similarity. They found effects of numerical size that were consistent with the results in Schwarz and Eiselt (2012) for targets other than the digit 5: search was more efficient when the numerical distance between the target and distractor digits was large than when numerical distance was small. By controlling for the visual feature of shape, the work of both Schwarz and Eiselt and Godwin et al. demonstrate a role for numerical distance in visual search that cannot be attributed to shape differences. Furthermore, numerical size is analogous to visual similarity, such that visual search efficiency increases with the target-distractor numerical distance.

Sobel et al. (2015) used a different method to control for shape differences. Instead of creating a two-dimensional map of the visual similarity of digits rendered in the font (i.e., Verdana) used in Schwarz & Eiselt (2012) and Godwin et al. (2014), they created digits from line segments as on the faces of digital clocks, depicted in Figure 1. Furthermore, the target digits in the first condition were the distractor digits in the second condition, and the distractor digits in the first condition were the target digits in the second condition. And finally, they used digits that were as similar as possible in shape (Cohen, 2009); the target digits 5 and 6 in the first condition only had one line segment difference between them (the lower left vertical line segment present in the 6 was absent in the 5), and the distractor digits 3 and 8 had two line segment differences (the left vertical segments present in the 8 was absent in the 3). Participants viewed displays that each contained one of the target digits and varying numbers of the distractor digits. When the target digits were 5 and 6 and distractors were 3 and 8, the target digits were next to each other on the number line and distractors were separated from each other; this was called the inner targets condition (see Figure 2 for target and distractor positions on the number line). When the target digits were 3 and 8 and the distractors were 5 and 6, the target digits were separated from each other and the distractor digits were next to each other on the number line; this was called the outer targets condition. Even though both conditions used the same four digits, Sobel et al. found that search was more efficient for the inner targets condition than for the outer targets condition. This indicates that participants were inclined to group targets by their number line position; when the target digits were beside each other in a number line, it allowed participants to create a simpler target set than when the target digits were separated by the distractor digits.

Figure 1
Digits rendered in a "digital clock" font to control shape differences as in Krause et al., 2017, and Sobel et al., 2016

Figure 2
Number line positions for targets and distractors in Sobel et al., 2016



While these three studies (Godwin et al., 2014; Schwarz & Eiselt, 2012; Sobel et al., 2015) demonstrated that number line position can affect visual search performance, all three can be interpreted as consistent with the classical cognition-perception divide (Firestone & Scholl, 2015; Pylyshyn, 1999; Wolfe & Horowitz, 2004) because number line position affects the cognitive aspects of visual search as described in the following argument. Hout & Goldinger (2015) argued that when searching for a target item, visual items are compared to the objects stored in a mental target template to identify whether the visual items are targets. These target templates are used more in everyday life than one might expect. When planning to make a salad, you might find yourself in the produce section of the grocery store. Looking at all of the produce, the specific ingredients needed for a salad would be held in your mind by envisioning lettuce, carrots, tomatoes, and cucumber as your eyes scan the shelves. When you examine some particular produce item (e.g., asparagus), you compare the visual image of the asparagus to each item in the target template, eventually deciding that it is not one of the target items. Thus, searching for salad ingredients is not just a search through the external visual landscape, but also proceeds through an internal target template. This kind of task, in which searches proceed through both the external visual field and the internal target template is called hybrid search.

For the studies in which the numerical distance between target and distractor digits was varied (Godwin et al., 2014; Schwarz & Eiselt, 2012), comparisons to the target template can be carried out more quickly when the target and selected item are numerically distant than

when numerically close (Moyer & Landauer, 1967). When target digits are adjacent to each other on the number line (Sobel et al., 2015), maintaining the target template in working memory requires less effort than when target digits are separated from each other, which frees up mental resources and allows for efficient comparisons between selected items and target digits. With the role for numerical distance in visual search firmly established, researchers turned to the question of whether the interaction between a target's numerical size and its physical size (i.e., font size) can affect visual search performance).

Size Congruity Effect

The interaction between a digit's numerical size and physical size has traditionally been investigated using size congruity experiments. In the original size congruity experiments, participants were asked to examine two different numbers and select the larger or smaller number out of the pair based on their numerical value (Besner & Coltheart, 1979). In these experiments, the physical size (i.e., font size) was also manipulated. For the numbers 9 and 2, for example, 9 would be considered numerically larger and 2 numerically smaller. The font size of the number 2 could be increased (2, 9) to make it physically larger than 9. Therefore, in some size congruity experiments participants are asked to select the physically larger number (Henik & Tzelgov, 1982). Because both numerical and physical size can be manipulated, the numerical and physical size can be either congruent or incongruent with each other. In congruent trials the numbers shown to participants were considered either large or small both in terms of physical size and numerical size. The number 9 for example, is numerically larger than 2, so in congruent trials it would be presented in a large font size as well. There were incongruent trials as well, where a large numerical size, such as 9 was expressed in a small font size. In a typical size congruity experiment, RTs are faster in the congruent trials, where numerical size and physical size are the same, than in incongruent trials.

Risko et al. (2013) sought to understand how attention might be implicated in size congruity effects. Participants viewed two numbers and indicated the location of the numerically larger number, as in typical numerical comparison tasks such as Besner and Coltheart (1979). In control trials, both numbers appeared at the same time, but in experimental trials either the physically smaller number or physically larger number was presented first, then 100 ms later the other number appeared. The authors predicted that when one digit appeared before the other (experimental trials), the first item to appear would grab attention. As expected, the size congruity effect, as indicated by the RT difference between congruent and incongruent trials, was greater when the physically larger item appeared first than when the two items appeared simultaneously. Apparently, the item's larger *physical* size grabbed participants' attention even though they were instructed to pay attention to *numerical* size. This result supports Risko et al.'s claim that visual attention is implicated in size congruity experiments. However, Arend and Henik (2015) identified a limitation in Risko et al.: their experimenters had only asked participants to locate the numerically larger target. Arend and Henik argued that if Risko et al. had instructed participants to localize not just the numerically larger target but also the numerically smaller target, the effect of attention might have disappeared.

Nevertheless, even with the limitation identified by Arend and Henik (2015), Risko et al.'s (2013) finding that the size congruity effect depends on attention suggests that the visual attention required for visual search (Wolfe, 1998) may be affected by size congruity. After all, although Risko et al. had intended their task to be a numerical comparison, it was analogous to a visual search in which participants searched through visual displays containing just two items to localize the single target item. Emboldened by the results from Risko et al., later researchers recast their task as a visual search task containing several items (Krause et al., 2017, Sobel et al., 2016). They predicted that the size congruity effect found in numerical comparison would also arise in visual search tasks. Further, they addressed Arend and Henik's critique by asking participants not just to localize the numerically larger target, but also the numerically smaller target. The results indicated that in both cases, when participants were asked to find the numerically large target or numerically small target, there was a size congruity effect for the visual search; RTs were slower when the target's numerical and physical size were incongruent than when they were congruent.

Sobel and Puri (2018) explained this size congruity effect in terms of the target template in visual short term memory. In their first experiment, participants were instructed to search for one of two target items: one group of participants searched for numerically small targets (2 or 3), and another group searched for numerically large targets (8 or 9). Displays consisted of one of the two target digits among varying quantities of distractor digits. In Experiment 1, RTs were faster when the target's numerical and physical size were congruent than when they were incongruent, replicating the previous studies (Krause et al., 2017; Sobel et al., 2016). As in these previous studies, the distractor digits

were interleaved within a block, so participants who searched for a numerically large target could never tell whether a given display would contain an 8 or a 9 and they would need to maintain a range of digits (i.e., greater than 5) in their target template.

In a follow-up experiment, the target digits were presented in blocks rather than randomly interleaved. Thus participants who searched for a numerically large target would see an 8 in every display in one block and a 9 in every display in the other block. Presenting target digits in blocks eliminated the size congruity effect; RTs for congruent and incongruent targets were not different. Sobel et al. argued that in the first experiment, when the target digits were randomly interleaved, participants needed to load their target template with a range of numerical values (e.g., greater than 5), but in the follow-up experiment, when the target digits were arranged in blocks, participants could load their target template with the *shape* of the single digit they would see in every trial in the block. Thus, in the first experiment the physical and numerical size of the target item had the opportunity to interact in the target template, but in the follow-up experiment, targets were stored in the target template in terms of their shape, *not* their numerical size, so there was no opportunity for the target's physical and numerical size to interact in the target template. These experiments suggest that the effect of size congruity in visual search occurs primarily in the target template that is stored in visual working memory. After one of the search items has been selected, it is compared to the target template to determine whether search should be terminated as successful, or it should continue by visually selecting another search item. In Experiment 1, when the target was defined by its relative numerical size (e.g., greater than 5), any selected items with an incongruent (e.g., small) physical size delayed this decision making process. As with the effect of number line position on visual search, size congruity affects aspects of visual search processing (e.g., working memory) that occur on the cognition side of the perception-cognition divide, rather than the aspects (e.g., visual selection) that occur on the perceptual side of the perception-cognition divide.

Stroop Effect

The size congruity effect is sometimes called the numerical Stroop effect (e.g., Ativan et al., 2024; Shichel & Goldfarb, 2022), to emphasize the interference between semantic (i.e., numerical size) and perceptual features (i.e., physical size) that were first demonstrated by Stroop's (1935) seminal experiments. The fact that interference between numerical size and physical size (i.e., size congruity effect) can influence visual search performance implies that the interference between a word's meaning and ink color (i.e., Stroop effect) can similarly affect search. In modern versions of the Stroop task, participants view color words on a computer screen that are written with pixels that are either congruent with the target word's meaning (e.g., "Red" written in red pixels) or incongruent (e.g., "Red" written in blue pixels). In a Stroop condition participants identify the target's pixel color, and in a reverse Stroop condition participants identify the target's meaning instead of its pixel color. The modern day Stroop capitalizes on the advancements of technology by using computers to capture trial-by-trial RTs, something the original Stroop study could not accomplish when RTs were measured with handheld timekeepers. The difference in RTs between incongruent and congruent trials are typically larger when participants attend to the target's color (Stroop) than when they attend to the target's meaning (reverse Stroop); in other words, the Stroop effect is typically larger than the reverse Stroop effect. This asymmetry is so common that it is often referred to as the classic Stroop asymmetry (Melara & Algom, 2003). With the asymmetry being so common, some researchers have tried to find out if the asymmetry can be inverted; that is, can the Stroop effect be smaller than the reverse Stroop effect?

The answer is yes: using a task he called a matching task, Durgin (2000) inverted the classical Stroop asymmetry. In his experiment participants were shown a color word on a computer display. The meaning of the target word ("Red", "Green", "Blue", or "Yellow") would either be congruent or incongruent with its pixel color. The target word was surrounded by four colored (red, green, blue, and yellow) patches. In the Stroop task participants were instructed to attend to the target's pixel color, and to move their cursor from the target word to the color patch that matched the target word's pixels. For example, if the target word was "Red" in blue pixels, the correct response would be to move their cursor to the blue patch. In the reverse Stroop task participants were instructed to move their cursor from the target word to the color patch that matched the target word's meaning. For the target word "Red" in blue pixels, the correct response would be to move their cursor to the red patch. The results from these Stroop and reverse Stroop experiments found an inverted asymmetry; the Stroop effect was smaller than the reverse Stroop effect. Durgin attributed the inversion of the classic asymmetry to the role of translation in such a matching task. Translation refers to the participants processing the meaning of the target word, and then translating it into a visual color that can be matched to one of the color patches surrounding the target in the display. Durgin claimed

that the reverse Stroop effect was larger than the Stroop effect because translation was required in the reverse Stroop condition but was not required in the Stroop condition. Durgin's inversion of the classic Stroop asymmetry has been replicated in other matching tasks (Miller et al., 2016; Song & Hakoda, 2015; Yamamoto et al., 2016).

Sobel et al. (2020) noted that what Durgin and others (Durgin, 2000; Miller et al., 2016; Song & Hakoda, 2015; Uleman & Reeves, 1971; Yamamoto et al., 2016) called a matching task used displays containing several items, and participants were instructed to select one of the display items, which is a kind of visual search, even though the authors didn't explicitly refer to their tasks as "visual search". Sobel et al. argued that traditional Stroop experiments required identification (i.e., identify the target's color or meaning), whereas these matching tasks required localization (i.e., indicate the location of an item defined by its color or meaning), and that identification primarily depends on verbal processing but localization primarily depends on visual processing. Thus, the simple fact that matching tasks require localization should be sufficient to invert the classical Stroop asymmetry from traditional experiments that require identification, even without any need for translation.

To support their argument that the distinction between identification and localization is sufficient to invert the Stroop asymmetry without the need for translation, Sobel et al. (2020) manipulated the task requirements and eliminated the need for translation. In Experiment 1 participants viewed target color words with congruent or incongruent pixel colors, and identified the pixel color of the target word (Stroop condition) in one block and the meaning of the word (reverse Stroop condition) in the other block. They identified the target's color or meaning by pressing one of four keys on the keyboard. The Stroop effect was larger than the reverse Stroop effect, which replicated the classic Stroop asymmetry. In Experiment 2 participants were presented with a display containing a cue in the center that was either a patch of color (Stroop condition) or a color word written in black pixels (reverse Stroop condition). Surrounding the cue were four color words ("Red", "Green", "Blue", or "Yellow") each written with pixels that were one of four colors (red, green, blue, or yellow), arranged on an imaginary circle centered on the cue. The search items' meanings and pixel colors were either all congruent or incongruent. Participants pressed one of two keys to indicate whether the target was on the right or left side of the display. Even though no translation was required, the visual processing required for localization was sufficient to invert the classic Stroop asymmetry: the Stroop effect was smaller than the reverse Stroop effect. Smith et. al (2022) showed that the inversion of the classical Stroop asymmetry generalizes to a diverse sample with experimental materials delivered remotely via the internet.

We have argued that even though semantic associations can influence visual search performance, all the effects discussed prior to the word-color Stroop effect (thorn targets among B, b, and p, numerical size and grouping on the number line, size congruity effect) are nevertheless consistent with the classical perception-cognition divide (Firestone & Scholl, 2015; Pylyshyn, 1999; Wolfe & Horowitz, 2004). We are inclined to believe the same about the effect of Stroop interference on visual search, but must admit that further research will be needed to definitively support such an interpretation. Nevertheless, we can imagine something along the following lines. Eimer (2015) argues that visual search proceeds in different processing stages: preparation, guidance, selection, and recognition. Visual processing is implicated in the guidance and selection stages, and working memory is implicated in the recognition stage. As with the previously discussed effects of semantic associations on visual search, we believe that Stroop incongruity affects the recognition stage of visual search, only after visual processing has selected one of the search items. Therefore, we believe that these results are consistent with the perception-cognition divide. Future research will determine whether we are correct.

Conclusion

Visual search performance has long been presumed to be driven by visual features such as color and shape. Early attempts to find a role for the semantic associations of search items such as numbers and letters were equivocal, and a repeated failure to replicate the earliest effect of alphanumeric categories on visual search had a chilling effect on further investigations for the next two decades. But Lupyan (2008) had the courage to take another crack at it, and the insight to design target and distractor items that allowed him to show a clear effect of letter identity on visual search efficiency. And while Lupyan seemed to argue that his results sounded the death knell for the classical perception-cognition divide, his results could also be interpreted in a manner that is consistent with the divide. He also opened the door for other researchers who found effects of numerical distance, number line grouping, and numerical size congruity on visual search. These studies that extended on Lupyan further supported the claim that semantic associations affect the role of cognition in visual search rather than basic perceptual processing itself. Recent studies have found that the classic Stroop asymmetry can

be inverted in visual search localization tasks, but further research is needed to demonstrate whether this effect is attributable to the cognitive or perceptual processing in visual search.

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